Howard University Data Science

Project 2

Topic: Volume vs Efficiency in the NBA 2024-25 Season

By

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Objective

Our objective with this project was to study the relationship between scoring volume and efficiency with the NBA 2024-25 season. Specifically, we aimed to investigate a player's shooting efficiency and how an increased volume of shooting can negatively impact a player's shooting efficiency. By using visual data, we can uncover how playing time, shot attempts, and team systems correlate with true scoring effectiveness.

Introduction

In the modern day, high velocity NBA the rise in shot volume has raised a question, at what point does increasing the amount a player shoot diminish the value/efficiency of the shot? While total points often are the main focus of headlines, Team Blockchain looked into the intersection of volume and efficiency during the 2024-25 season data. Through analyzing Point Per Shoot (PPS) we can distinguish between high-volume shooters and elite offensive anchors.

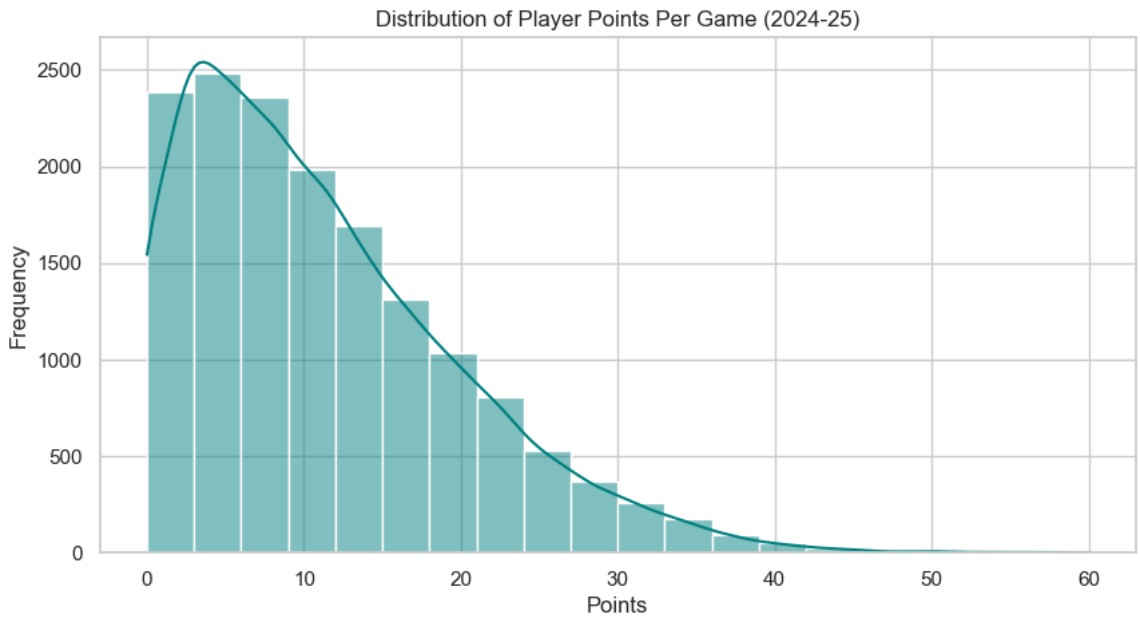
Method

For our project we used a comprehensive dataset of NBA player performances from the 2024-25 NBA season. To begin, we imported all the data into Jupyter Notebook and created a PPS metric. This was so that we could measure scoring efficiency. We did some data cleaning by filtering out entries with zero field goal attempts to prevent any mathematical errors. For section II, we grouped individual game logs into season long totals for each player. We put a minimum 20 field goal attempt threshold just to clean up

any outliers/noise from the low-minute players. And we used Seabron/Matplotlib to explore the data visually.

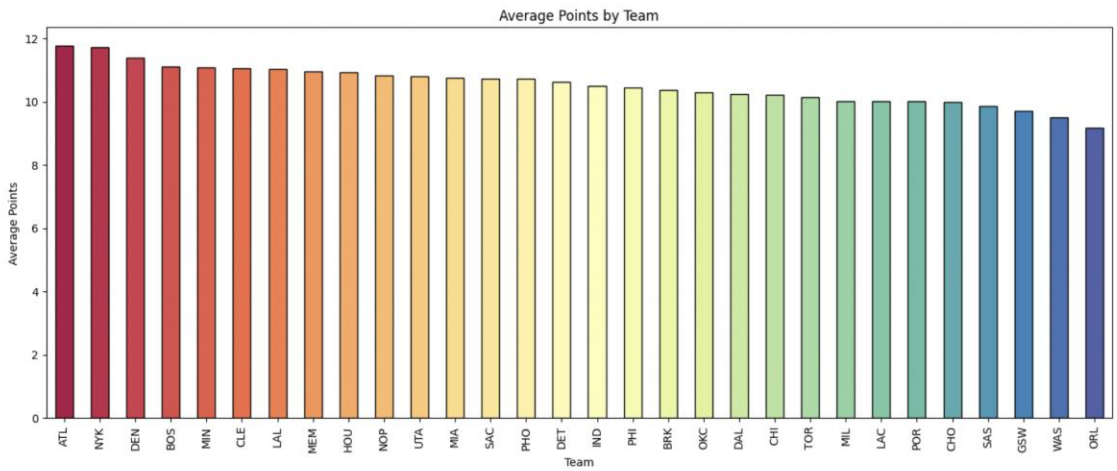
Storytelling

Graph 1 Histogram (Distribution of Points):



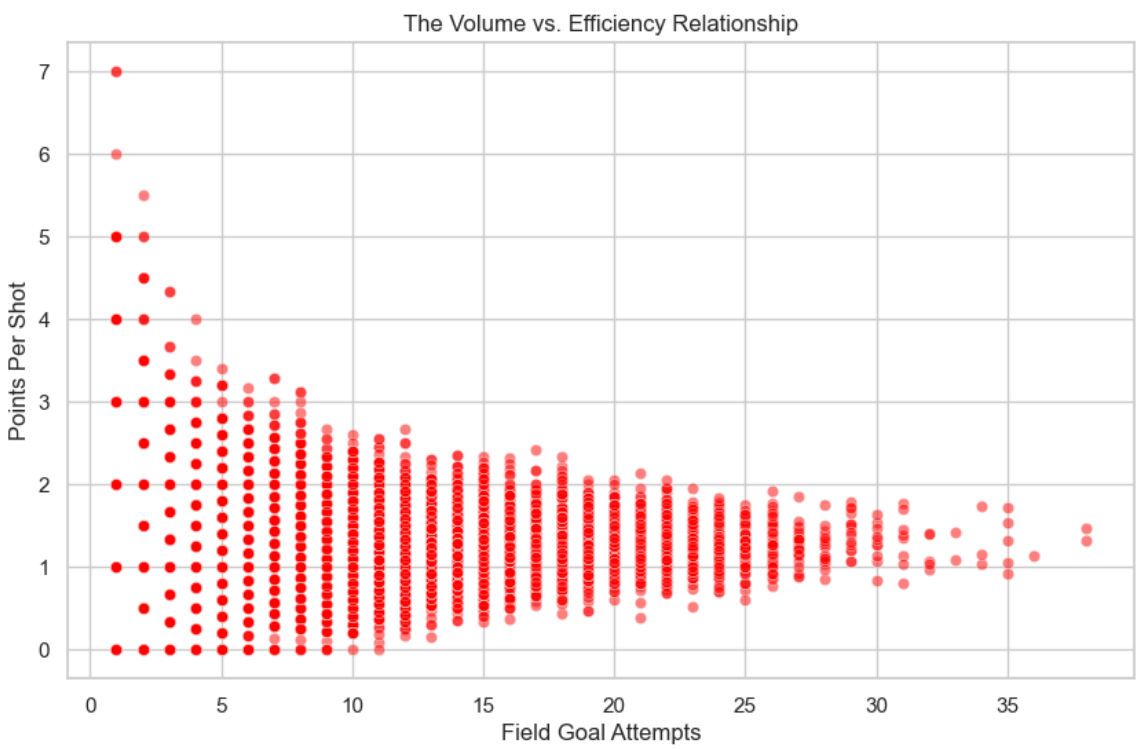
Explanation: This graph shows that most NBA players score under 10 points a game and that players that score very high are few.

Graph 2 Bar Chart (Average Efficiency by Team):



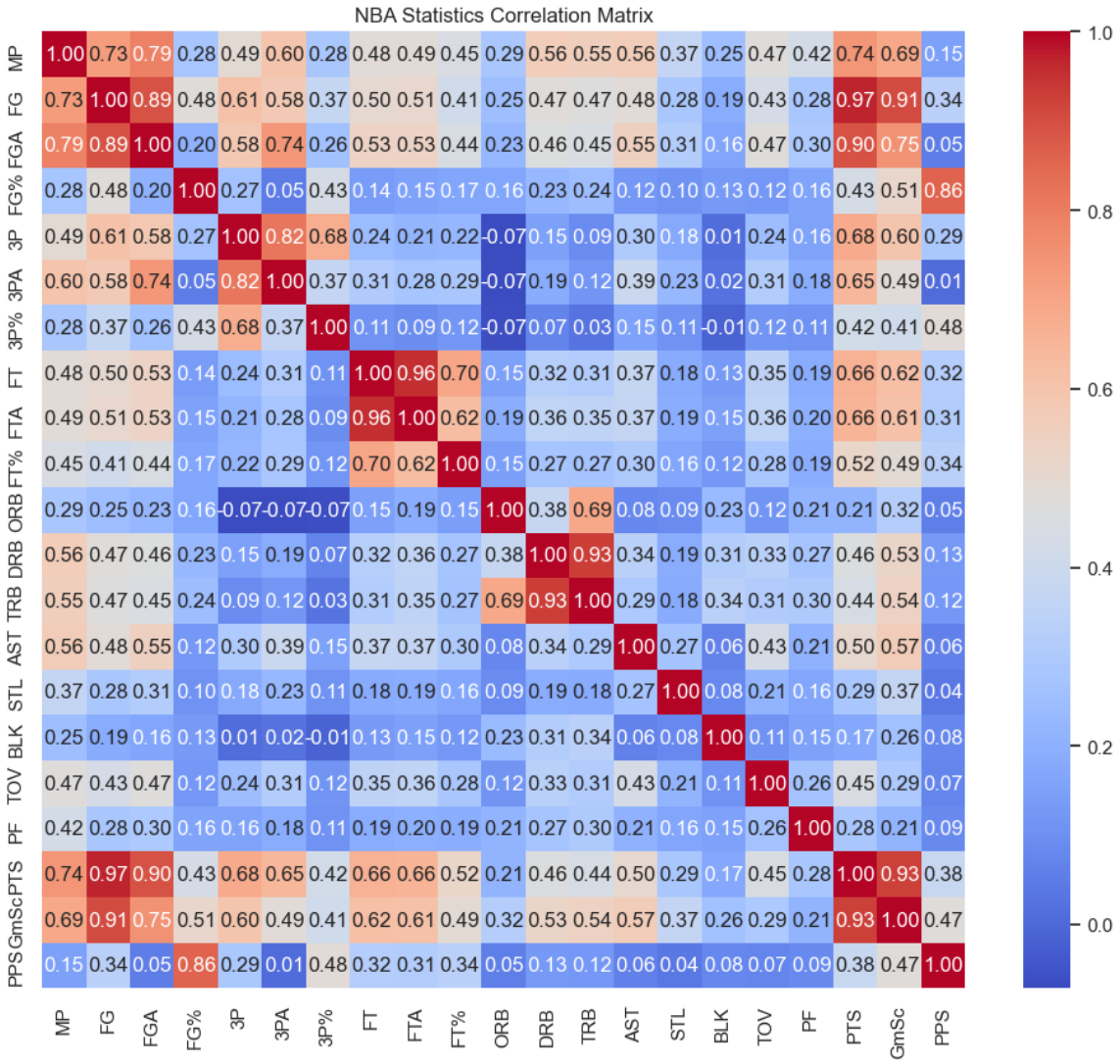
Explanation: Teams like Memphis and Clevland lead the pack in PPS which means they take the highest quality shots.

Graph 3 Scatter Plot (Volume vs. Efficiency):



Explanation: This graph shows that as shots go up efficiency goes down for most players.

Graph 4 Heatmap (Correlation Matrix):



Explanation: There is a correlation between minutes played and points, but a much weaker correlation between Field Goal Attempts and Field Goal Percentage.

Conclusion

In conclusion, our visual analysis of the 2024-25 NBA season reveals that high-volume scoring typically comes at the expense of efficiency. While playing more minutes naturally leads to more total points, our data shows that as a player takes more shots, their

PPS and overall field goal percentage generally decrease. The rarity of elite, high-efficiency scorers emphasizes that sheer shot volume does not guarantee offensive success. Instead, modern basketball relies heavily on shot quality and strong offensive systems, like those seen in Memphis and Cleveland, to maintain high scoring efficiency.

References

NBA Stats (2024-2025 Season). *Player Game Logs and Advanced Metrics.* Retrieved from Kaggle / Basketball-Reference via https://www.kaggle.com/datasets/eduardopalmieri/nba-player-stats-season-2425?select=database_24_25.csv

Acknowledgements

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